Simplifying Software Metric Models via Hierarchical LASSO with Incomplete Data Samples

Tianfa Xie
College of Applied Sciences
Beijing University of Technology
Beijing 100124, P.R. China
e-mail: xtianfa@amss.ac.cn

W. Eric Wong
Department of Computer Science
University of Texas at Dallas
Richardson, TX 75083, U.S.A.
e-mail: ewong@utdallas.edu

Wenxing Ding
China National Institute of Standardization
4 Zhichun Road, Haidian District,
Beijing 100088, P.R. China
e-mail: dingwx@cnis.gov.cn

***Abstract*—Software metric models can be used in predicting the interested target software metric(s) for future software project based on certain related metric(s). However, during the construction of such a model, incomplete data often appear in data sample gained from analogous past projects. In addition, whether a particular continuous predictor metric or a particular category for a certain categorical predictor metric should be included in the model must be determined in practice. To solve these problems, this paper introduces a methodology integrating the k-nearest neighbors (k-NN) multiple imputation method, kernel smoothing, Monte Carlo simulation, and a latest variable selection method. Thus, a more flexible model is constructed. A case study is given to illustrate the proposed procedures.**

Keywords-software metrics; model simplification; missing data; multiple imputation; variable selection

I. INTRODUCTION

Software metrics are measurements or indications of some properties of a software system, its specifications, and/or the software project that developed it. Typical examples are the program size of a system (e.g., in number of lines of code), the number of defects in a system found during initial live-run, the project team size, the project work effort (e.g., in number of person-hours), etc.[1].

To explore the possible relationship among these metrics, appropriate models have been constructed. All such models are known as software metric models, although other terminologies have also been used. In general, each model gives a relationship between a specific target metric (a dependent variable) and one or more predictor metrics (independent variables).

Construction of software metric models invariably makes use of data samples of metrics from analogous past projects. Some problems will be encountered, which include incomplete data in data sample, variable selection and level collapse during model construction. Details of such problems are described in [2]. To cope with these problems, maximum likelihood (ML) method was developed [3], [4], and stepwise regression along with other techniques was also used [2]. Unfortunately, the ML methods assuming multivariate normal distribution of data, which may not be the case for software metric data samples [4]. Meanwhile, although stepwise deletion and other subset selection procedures are practically useful to select significant variables, these selection procedures ignore stochastic errors inherited in the stages of variable selections. Hence, their theoretical properties are somewhat hard to understand. Furthermore, the best subset variable selection suffers from several drawbacks, and the most severe one is its lack of stability [5]. Recent study shows that regularized methods such as BRIDGE, LASSO and SCAD, are much more favorable in variable selection [6]. They do estimation and variable selection at the same time through some constrained optimization criteria. In theory, such method can obtain the true model provided that it was known, which is called Oracle property [6]. Thus we will take one of such variable selection approaches instead of stepwise regression in this paper.

As in [2], this paper adopts dummy variables to represent categorical predictors. In that way, level combination or collapse for categorical predictors turns into dummy variable selection. Thus, both continuous predictor selection and categorical level combination can be treated simultaneously under the variable selection framework. Moreover, previous work shows that linear regression often lacks of fit, in other words, produce large prediction error [2]. Scatter plot also indicates that there may be nonlinear relations between target metric and predictors. It is a good way to include high order polynomial terms of continuous predictors in the regression model, because such models make linear regression model as its sub-model. In the following, variable selection for both continuous and categorical predictors is properly implemented via adaptive hierarchical LASSO which will be addressed in detail in section II.C.

The remainder of this paper is organized as follows. The new methodology is explained in Section II. Focus goes to hierarchical LASSO (HLASSO). A case study to illustrate the applicability of this new methodology is given in Section III. Comparison with stepwise regression is also given. Finally, the conclusion appears in Section IV.

II. THE NEW METHODOLOGY

Basically, the new methodology of this paper uses 'linear' regression analysis to determine the relationship of a target metric (or dependent variable) with one or more predictor metrics (or independent variables). The general form of such a model is at least initially a regression equation as follows:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \beta_{p+1} x_{p+1} + \cdots + \beta_q x_q + \varepsilon \qquad (1)$$

978-0-7695-4303-1/10 $26.00 © 2010 IEEE
DOI 10.1109/WCSE.2010.117

where y is the target metric,

$x_1, \ldots, x_p$ are the continuous predictor metrics,

$x_{p+1}, \ldots, x_q$ are the categorical predictor metrics that take on values of discrete levels, for example, the categorical predictor metric "development type" taking on values 1, 2, and 3 which correspond to "enhancement," "new development," and "redevelopment," respectively,

$\beta_0, \beta_1, \ldots, \beta_q$ are the regression coefficients to be estimated during the model construction, and

ε is the error term.

A. The k-NN Multiple Imputation

Since regression model needs complete data, incomplete data in data samples that are used to construct software metric models should first be tackled. The same k-NN multiple imputation method is implemented as [7].

By independently sampling from estimated density function for any missing continuous metric in any incomplete observation and by independently sampling from estimated distribution function for any missing categorical metric in any incomplete observation, a Monte Carlo simulation is actually done to generate a large number of, say, N "completed" data samples. These N "completed" data samples are the same as the original data sample but with the missing metrics filled in through such sampling. For each of these N "completed" data samples, HLASSO variable selection is applied to simplify (1) by omitting redundant predictor metric(s) and combining categories for categorical predictor metrics. The details of such variable selection method are left to Section II.C.

B. Most Probable Multinomial Event Selection and the Sequential Procedure

Now that the new methodology is to generate N different "completed" data samples to which HLASSO is applied, such data samples could result in probably different and conflicting selection results, each being characterized by omitting different redundant predictor metric(s) and combining different categories for the categorical predictor metrics. The problem now is how to select the most possibly optimal result out of, say, m different HLASSO selection results. This can be solved by most probable multinomial event selection procedure. Details of the procedure are omitted here. Interested readers please refer to [2].

C. Variable Selection by HLASSO

Prior to the application of variable selection to each of the N "completed" data samples generated by the k-NN multiple imputation method, dummy variables are introduced into (1) to replace the categorical predictor metrics therein. For example, if categorical predictor metric x_{p+1} in (1) actually represents the categorical predictor metric "development type" such that x_{p+1} = 1, 2, and 3 when the "development type" is "enhancement," "new development," and "redevelopment" respectively, then this categorical predictor metric is said to have three levels given its being able to take on either one of these three values. Two dummy variables can thus be introduced for this 3-level categorical predictor metric as follows:

$$x_{p+1,1} = \begin{cases} 1 & \text{if } x_{p+1} = 2 \\ 0 & \text{otherwise} \end{cases}$$

$$x_{p+1,2} = \begin{cases} 1 & \text{if } x_{p+1} = 3 \\ 0 & \text{otherwise} \end{cases}$$

In other words, $(x_{p+1,1}, x_{p+1,2}) = (0,0)$ if $x_{p+1} = 1$, which is often referred to as the base level. Assuming that in general, categorical predictor metric x_j takes on values (or levels) 1, 2, …, n_j for $j = p + 1, p + 2, \ldots, q$, that is to say, x_j has n_j levels, by introducing $n_j - 1$ dummy variables for x_j , (1) is reformulated as the following regression equation:

$$\begin{aligned} y = {} & \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p \\ & + \beta_{p+1,1} x_{p+1,1} + \cdots + \beta_{p+1,n_{p+1}-1} x_{p+1,n_{p+1}-1} \\ & + \cdots \\ & + \beta_{q,1} x_{q,1} + \cdots + \beta_{q,n_q-1} x_{q,n_q-1} + \varepsilon \end{aligned} \tag{2}$$

Where $x_{j,k} = \begin{cases} 1 & \text{if } x_j = k+1 \\ 0 & \text{otherwise} \end{cases}$ for $j = p + 1, p + 2, \ldots, q,$

$k = 1, 2, \ldots, n_j - 1$ and $\beta_{p+1,1}, \cdots, \beta_{p+1,n_{p+1}-1}, \cdots, \beta_{q,1}, \cdots, \beta_{q,n_q-1}$ are the new regression coefficients to be estimated during the model construction. For certain continuous predictor metrics x_l for l = 1, 2, …, p or dummy variables $x_{j,k}$ for $j = p + 1, p + 2, \ldots, q$ and $k = 1, 2, \ldots, n_j - 1$ may not matter to the target metric y with sufficiently high statistical significance, the equation (2) can be simplified. If that happens to continuous predictor metric x_l for $l = 1, 2, \ldots, p$, then x_l should be omitted from (2) . If that happens to dummy variable $x_{j,k}$ for $j = p + 1, p + 2, \ldots, q$ and $k = 1, 2, \ldots, n_j - 1$, this means that categorical variable x_j taking on the value (or level) $k + 1$ is not distinguishable from its taking on the value (or level) 1 or its base level, and thus these two levels can be combined. In this way, (2) can be simplified to some extent.

Note that in regression equations (2), some of the explanatory variables are factors where each factor may be represented by a group of dummy variables. The dummy variables associated with a factor are called as grouped variable. $(x_{p+1,1}, \cdots, x_{p+1,n_{p+1}-1})$ and $(x_1, x_1^2, \cdots, x_1^d)$ are examples of grouped variable. For the regression with group predictors such as in ANOVA or the additive model with polynomial components, several recent papers have considered selecting important groups of variables using penalized methods, see [8] and references therein. These papers only considered group selection, but did not address the question of individual selection within groups. In our regression model, we also want to select the variable within the groups, especially for the dummy variables representing categorical predictors. Fortunately, to our knowledge, Huang et al. proposed group BRIDGE [9] and Zhou and Zhu [10] developed Hierarchical LASSO to tackle this issue. There are some similarities between these two papers. For ease of understanding and implement, HLASSO is introduced in detail and used afterward.

We assume that the prediction variables can be divided into K groups and the k-th group contains p_k variables. When $p_k = 1, k = 1, 2, \ldots, K$, it is to say that there is no group structure in prediction metrics. Specifically, the linear model (2) is now written as

$$y = \beta_0 + \sum_{k=1}^{K} \sum_{j=1}^{p_k} \beta_{kj} x_{kj} + \varepsilon \qquad (3)$$

where $(x_{11}, \ldots, x_{1p_1}), (x_{21}, \ldots, x_{2p_2}), \ldots, (x_{K1}, \ldots, x_{Kp_K})$ are all K group variables which may stand for a single variable or a single variable with its polynomial terms to certain order or dummy variables corresponding to a categorical predictor. We are interested in finding out which variables, especially which "groups," have an important effect on the response. If we find that k-th group is important, we are further interested in which variables in this group are important.

For equation (3), reparamterize β_{kj} as

$$\beta_{kj} = d_k \alpha_{kj},\ k = 1, \ldots, K;\ j = 1, \ldots, p_k$$

where $d_k \geq 0$. This decomposition reflects the information that β_{kj}, $j = 1, \ldots, p_k$ all belong to the k-th group, by treating each β_{kj} hierarchically. d_k is at the first level of the hierarchy, controlling β_{kj}, $j = 1, \ldots, p_k$, as a group; α_{kj} 's are at the second level of the hierarchy, reflecting differences within the k-th group. For the purpose of variable selection, the following penalized least squares criterion is considered:

$$\begin{aligned} \max_{d_k, \alpha_{kj}} \quad & -\frac{1}{2} \sum_{i=1}^{n} (y_i - \sum_{k=1}^{K} d_k \sum_{j=1}^{p_k} \alpha_{kj} x_{i,kj})^2 \\ & - \lambda_1 \sum_{k=1}^{K} d_k - \lambda_2 \sum_{k=1}^{K} \sum_{j=1}^{p_k} |\alpha_{kj}|, \qquad (4) \\ \textit{subject to} \quad & d_k \geq 0, k = 1, \ldots, K, \end{aligned}$$

Where $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ are tuning parameters. λ_1 controls the estimates at the group level, and it can effectively remove unimportant groups: if d_k is shrunken to zero, all β_{kj} in the k-th group will be equal to zero. λ_2 controls the estimates at the variable-specific level: if d_k is not equal to zero, some of the α_{kj} hence some of the β_{kj}, $j = 1, \ldots, p_k$, still have the possibility of being zero; in this sense, the hierarchical penalty keeps the flexibility of the $L1$-norm penalty. It is proved that that the two tuning parameters λ_1 and λ_2 in (4) can be simplified into one. Specifically, let $\lambda = \lambda_1 \cdot \lambda_2$, (4) is equivalent to

$$\begin{aligned} \max_{d_k, \alpha_{kj}} \quad & -\frac{1}{2} \sum_{i=1}^{n} (y_i - \sum_{k=1}^{K} d_k \sum_{j=1}^{p_k} \alpha_{kj} x_{i,kj})^2 \\ & - \sum_{k=1}^{K} d_k - \lambda \sum_{k=1}^{K} \sum_{j=1}^{p_k} |\alpha_{kj}|, \qquad (5) \\ \textit{subject to} \quad & d_k \geq 0, k = 1, \ldots, K, \end{aligned}$$

To estimate the d_k and α_{kj} in (5), the iterative algorithm is given in [10]. The adaptive idea in [11] can be used to improve the HLASSO method. For choosing the tuning parameter λ in (5), cross validation is often used. Thanks professor Zhu, adaptive HLASSO is readily implemented by his codes written by free statistical software R [12].

III. A CASE STUDY

The data sample used in this case study is the "Repository Data Disk-Release 9" which is from the International Software Benchmarking Standards Group (ISBSG). It is recognized as one of the most comprehensive, reliable, systematic and representative data samples of its kind [13], [14], [15]. This data sample consists of 100 metrics for a collection of 3024 software projects. However, most of these metrics suffer from incomplete data for some of the projects.

The data sample was used to construct a software quality metric model. Table 1 gives the details of the target metric y and all predictor metrics $x_1, x_2, \ldots, x_5$ included in this software quality metric model which initially is in the form of (1). The five predictor metrics $x_1, x_2, \ldots, x_5$ were initially selected based on [15]. Please note that in order to demonstrate the usefulness of the new methodology of this paper, this case study added the predictor metrics x_3 which, from the authors' experience-based intuition, is evidently redundant for models of similar purposes.

TABLE I. THE TARGET AND PREDICTOR METRICS OF THE SOFTWARE QUALITY METRIC MODEL

	Name of Metric	Description
y	Extreme defects	Extreme defects reported in the first month of software use. The only target metric or dependent variable.
x_1	Major defects	Major defects reported in the first month of software use.
x_2	Minor defects	Minor defects reported in the first month of software use.
x_3	User base – locations	Number of physical locations being served by the installed software system.
x_4	Function points	The adjusted function point count.
x_5	Development type	Specifies whether the development is an enhancement, new development, or redevelopment. Three categories: 1 for "enhancement," 2 for "new development," and 3 for " redevelopment"

In order to make the result more reliable, the data sample was preprocessed as in [2]. Thus, there were only 669 observations left.

Therefore, corresponding to (5), the new methodology reformulates the following regression equation.

$$
\begin{aligned}
y = \beta_0 &+ \beta_{1,1}x_{1,1} + \beta_{1,2}x_{1,2} + \beta_{1,3}x_{1,3} \\
&+ \beta_{2,1}x_{2,1} + \beta_{2,2}x_{2,2} + \beta_{2,3}x_{2,3} \\
&+ \beta_{3,1}x_{3,1} + \beta_{3,2}x_{3,2} + \beta_{3,3}x_{3,3} \\
&+ \beta_{4,1}x_{4,1} + \beta_{4,2}x_{4,2} + \beta_{4,3}x_{4,3} \\
&+ \beta_{5,1}x_{5,1} + \beta_{5,2}x_{5,2} + \varepsilon
\end{aligned} \tag{6}
$$

where $x_{i,j} = x_i^j,\ i = 1,\cdots,4;\ j = 1,2,3$,

$$x_{5,1} = \begin{cases} 1 & \text{if } x_5 = 2 \\ 0 & \text{otherwise} \end{cases} \text{ and } x_{5,2} = \begin{cases} 1 & \text{if } x_5 = 3 \\ 0 & \text{otherwise} \end{cases}.$$

Stepwise regression and HLASSO variable selection are used respectively. For both methods, we set $k = 5$ and use the Epanechnikov kernel function [16] in the k-NN density estimation, and the bandwidth $h = 5^{-0.2} = 0.72478$. The values of P^* and θ^* for the sequential procedure [17] were set to 0.99 and 1.005 respectively which guarantee a confidence level of correct selection at 99% which is practically high enough. Likewise, $\theta^* = 1.005$ is practically small enough. Upon generating $N = 2000$ "completed" and using the notations as [2], then $Y_{[m]N}$ to $Y_{[m-3]N}$ are shown in the following Table II.

TABLE II. $Y_{[M]N}$ TO $Y_{[M-3]N}$ FOR THE SOFTWARE QUALITY METRIC MODEL

	$Y_{[m]N}$	$Y_{[m-1]N}$	$Y_{[m-2]N}$	$Y_{[m-3]N}$
Stepwise	799	1	0	0
HLASSO	748	29	15	7

The following summarizes the selection results of stepwise regression and HLASSO that corresponds to $Y_{[m]N}$ in Table II. Common result is

- The continuous predictor metrics "User base – location" (x_3) should be omitted.
- The categories "enhancement" (1) and "new development" (2) for the categorical predictor metric "development type" (x_5) should be combined.

The difference is for "Function points" (x_4), in stepwise all three terms are reserved while for HLASSO only second order is retained.

We also report the comparison results for different selected models by estimated mean square error (MSE) and mean relative error (MRE) through K-fold Cross Validation.

TABLE III. ERRORS FOR FINAL MODEL GAINED BY STEPWISE OR HLASSO

Type of error	MSE	MRE
Stepwise	73.86343	522744.1
HLASSO	71.50511	432703.1

MRE is much large here for both methods. This may lies in that for those dependent metric "Extreme defects" whose true values are zero, a very small constant say 10^{-6} was used as denominator instead of zero.

IV. CONCLUSION

This paper introduces a more flexible model and integrates a more advanced variable selection method. The case study shows that it performs better than stepwise regression. This also highlights the usefulness of this new methodology.

ACKNOWLEDGMENTS

We greatly appreciate Professor Victor K. Y. Chan's insightful ideas and financial support from the Science and Technology Development Fund of the Government of the Macao Special Administrative Region (No. 045/2005/A), Specialized Research Fund for the Doctoral Program of Higher Education (No. 20091103120012) and National Natural Science Foundation of China (No. 10971007).

REFERENCES

[1] Y. Zhou and H. Leung, "Empirical Analysis of Object-Oriented Design Metrics for Predicting High and Low Severity Faults," *IEEE Trans. Software Eng.*, vol. 32, no. 10, pp. 771-789, Oct. 2006.

[2] Tianfa Xie and W. Eric Wong, An Improved Method to Simplify Metric Models Constructed with Incomplete Data Samples, IEEE computer society, Fuzzy Systems and knowledge discovery 2010,1682-1687.

[3] I. Myrtveit, E. Stensrud, and U.H. Olsson, "Analyzing Data Sets with Missing Data: An Empirical Evaluation of Imputation Methods and Likelihood-Based Methods," IEEE Trans. Software Eng., vol. 27, no. 11, pp. 999-1013, Nov. 2001.

[4] V. Chan and W. E. Wong, "Optimizing and Simplifying Software Metric Models Constructed Using Maximum Likelihood Methods," *Proc. 29th IEEE Int'l Computer Software and Applications Conf.*, pp. 65-70, Edinburgh, UK, Jul. 2005.

[5] Breiman, L. (1996), "Heuristics of Instability and Stabilization in Model Selection," The Annals of Statistics, 24, 2350–2383.

[6] Fan, J., and Li, R. (2001), "Variable Selection via Nonconcave Penalized Likelihood and Its Oracle Properties," Journal of the American Statistical Association, 96, 1348–1360.

[7] Victor K.Y. Chan, W. Eric Wong, and T.F. Xie, "A Statistical Methodology to Simplify Software Metric Models Constructed Using Incomplete Data Samples: Its Procedure and Two Case Studies," International Journal of Software Engineering and Knowledge Engineering, vol. 17, no. 6, pp. 689-707, 2007.

[8] Zhao, P, Rocha, G. and Yu, B. (2006). Grouped and hierarchical model selection through composite absolute penalties. Annals of Statistics (to appear).

[9] Jian Huang et. al. "A group bridge approach for variable selection" , *Biometrika* 2009 96(2):339-355

[10] Zhou, N. and Zhu, J. (2007) Group variable selection via a hierarchical lasso and its oracle property. Unpublished manuscript. http://arxiv.org/abs/1006.2871

[11] Zou, H. (2006), "The Adaptive Lasso and Its Oracle Properties," *Journal of the American Statistical Association*, 101, 1418–1429.

[12] W.N. Venables and B.D. Ripley, Modern Applied Statistics with S. New York: Springer, 2002.

[13] I. Myrtveit, E. Stensrud, and U.H. Olsson, "Assessing the Benefits of Imputing ERP projects with Missing Data," Proc. 7th IEEE Int'l Software Metrics Symp., pp. 78-84, London, UK, Apr. 2001.

[14] R. Jeffery, M. Ruhe, and I. Wieczorek, "Using Public Domain Metrics to Estimate Software Development Effort," Proc. 7th IEEE Int'l Software Metrics Symp., pp16-27, London, UK, April 2001.

[15] B.J. Milne and K.B.G. Luxford, Worldwide Software Development: the Benchmark, Release 5. Warrandyte, Australia: International Software Benchmarking Standards Group Limited, 1998.

[16] J. Simonoff, Smoothing Methods in Statistics. New York: Spring-Verlag, 1996.

[17] D. Goldsman, S.-H. Kim, and B.L. Nelson, "Statistical Selection of the Best System," Proc. 2005 Winter Simulation Conf., pp. 178-187, M.E. Kuhl, N.M. Steiger, F.B. Armstrong, and J.A. Joines, eds,2005.